

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

Detailed guide and project requirements for the Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor analysis.

Background and Context:

Nassau Candy Distributor operates as a national distributor, shipping products from factories to customers across multiple US regions. In such operations:

- Shipping efficiency directly affects customer satisfaction
- Delays increase operational cost
- Inefficient routes reduce scalability

Despite having rich order and shipment data, logistics decisions are often made without route-level efficiency intelligence.

Problem Statement

The organization currently lacks clarity on:

- Which factory-to-customer routes are consistently efficient
- Which routes experience frequent delays
- How shipping performance varies by region, state, and ship mode
- Where operational bottlenecks exist geographically
- Without this visibility, logistics optimization remains reactive rather than data-driven.

Field	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date of order
Ship Date	Date of shipment
Ship Mode	Shipping method of order
Customer ID	Unique customer identifier
Country/Region	Country or region of customer
City	City of customer
State/Province	State/province of customer
Postal Code	Postal code / zip code of customer
Division	Product division
Region	Region of customer
Product ID	Unique product identifier
Product Name	Product long namer
Sales	Total sales value of order
Units	Total units of order
Gross Profit	Gross profit of order (Sales - Cost)
Cost	Cost to manufacture

Analytical Methodology (Step-by-Step)

Data Cleaning & Validation

- Validate date formats
- Remove invalid or negative lead times
- Handle missing shipment records
- Standardize geographic fields

Feature Engineering

- Calculate Shipping Lead Time (days)
- Categorize routes by:
 - Factory → Customer Region
 - Factory → Customer State
- Group shipments by Ship Mode

Route Definition & Aggregation

Each route is defined as: Factory Location → Customer State / Region For each route:

- Total shipments
- Average shipping lead time
- Lead time variability

Efficiency Benchmarking

- Rank routes from fastest to slowest
- Identify:
 - Top 10 most efficient routes
 - Bottom 10 least efficient routes
- Compare performance across ship modes

Geographic Bottleneck Analysis

- Identify regions with:
 - High average lead time
 - High shipment volume + poor performance
- Detect congestion-prone states or regions

Ship Mode Performance Analysis

- Compare shipping efficiency by:
 - Standard shipping
 - Expedited shipping
- Evaluate cost-time tradeoffs (descriptive)

KPI	Description
Shipping Lead Time	Ship Date – Order Date
Average Lead Time	Mean shipping duration per route
Route Volume	Number of orders per route
Delay Frequency	% of shipments exceeding threshold
Route Efficiency Score	Normalized lead-time performance

Streamlit Web Application Requirements

Dashboard Modules

- Route Efficiency Overview
 - Average lead time by route
 - Route performance leaderboard
- Geographic Shipping Map
 - US heatmap of shipping efficiency
 - Regional bottleneck visualization
- Ship Mode Comparison
 - Lead time comparison by shipping method
- Route Drill-Down
 - State-level performance insights
 - Order-level shipment timelines

User Capabilities

- Date range filter
- Region / State selector
- Ship mode filter
- Lead-time threshold slider

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

Conclusion

This project establishes a clear, data-driven understanding of shipping route efficiency for Nassau Candy Distributor. By transforming raw order and shipment data into route-level operational intelligence, the organization gains actionable insights to improve logistics performance, reduce delays, and enhance nationwide delivery reliability.